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AgFlow User's Manual

Version v6.00

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For use beginning with
TopAGNPS v6.00.a.000

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AgFlow User's Manual:

Introduction:

“AgFlow” is a module in the suite of modules called TopAGNPS. Its purpose is to calculate the hydraulic and hydrologic parameters for cells and reaches defined by TopAGNPS including the average RUSLE topographic factor (LS-factor) for each AnnAGNPS cell.

Most of the input files for AgFlow, as described below, can be generated by executing the following modules: DEDNM, RASPRO, RASFOR. The TopAGNPS modules, (DEDNM, RASPRO, and RASFOR), produces spatial files required by AgFlow. AgFlow produces a spatial and tabular output files. These files are used as input to all of the subsequent modules within TopAGNPS.

AgFlow produces several output files which are described below.

Input files:

There are several input files, described below, necessary for the successful execution of AgFlow. The first is an optional control file and the remaining are required spatial layer files. The layer files are specifically named as described below.

Control files:

AGFLOW.CSV (optional)

The optional csv-formatted control file allows the user to specify two keywords that affect the processing of spatial data and the determination of AnnAGNPS cell and reach delineations.

“ASC_Path”:

This optional keyword allows the user to specify the path to where the required spatial files created by DEDNM, RASPRO, and RASFOR are located.

This keyword is optional and if not included then the default location of the spatial input files created by DEDNM, RASPRO, and RASFOR will be assumed. The default relative path is the folder where TopAGNPS was invoked. If the keyword is present with a path specified, then that path will be used for all of the required spatial input files. If the keyword is used but its value is blank, then the current default path from which TopAGNPS is executed will be used for all of the spatial input files.

NOTE: This keyword is only used for independent runs of the AGFLOW module. It is ignored when AGFLOW is run in sequence with DEDNM, RASPRO, and RASFOR. In this case the output files from AGFLOW will be placed in the default folder location.

“REASSIGN”:

This optional keyword allows for a new feature for reassigning subareas or AnnAGNPS cells based on raster flows into each reach. Default=0, do not reassign subareas.

The “REASSIGN” option needs to be set to “1” indicating that this feature is to be invoked.

AgFlow reads the required input (.ASC) files including the SUBWTA.ASC. Right after the input files are read in, AgFlow then examines each reach and finds all the rasters that flow directly into the reach and stores that information. Then processing from the upstream end of the reach to the downstream end of the reach, each stored non-reach raster is traversed upstream to create unique subareas flowing into the reach.

New unique cell IDs are assigned and a new file, “SUBWTA_REASSIGNED.ASC” is written to output for verification purposes. The new cell IDs will contain the original cell ID * a cell multiplier that is calculated based on the maximum number of rasters found the longest reach path * 10 + (1 or 2 or 3 or 4) based on the source, left, right, or reach designation.

For example: Using a cell ID of “23”. Assume the reach with the largest number of rasters in its path is 83. In this case, 2 digits are needed to represent the reach raster location – or 10^2 . Assume that we’re at the upstream end of the reach path, the raster position would be 1. The new unique cell ID would be:

$(\text{Cell id} * \text{multiplier} + \text{rch_ras pos}) * \text{digit for src,l,r,rch} + \text{src,l,r,rch value} = \text{new cell ID}$
 $(23 * 100 + 1) * 10 + 3 = 23013$

Once the process has completed, AgFlow continues processing as it normally would.

This will expectedly create many new AnnAGNPS cells. Whereas, before, AgFlow only produced one cell for the source, one for the left and one for the right side, now it will produce one cell for the source but potentially multiple left cells and multiple right cells flowing into the reach. Each new cell is a sub-catchment within the original AnnAGNPS cell.

AGFCNT.INP (optional)

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The optional fixed-formatted control file allows the user to specify a number of parameters that affect the processing of spatial data and the determination of AnnAGNPS cell and reach delineations. If "AGFCNT.INP" is not used, defaults will be supplied for each parameter.

"AGFCNT.INP" is a fixed-formatted control file that takes the following form:

Line No.	Columns 1-80								
1	VERSION 3.10								
2	MISSION CREEK WATERSHED - OREGON								
3	Code 1: 200 & 500 Code 2: 26 & 200 Code 3: 10 & 100 Code 4: 10 & 50								
4	Using default values for LS-factor parameters								
5	.\								
6	.\								
7	.\								
8	LS-Factor Parameters						Sub_Wta		Optimize
9	Slope Code (1, 2, or 3)	Max. Drainage Area (ha)	Max. Profile Length (m)	Max. Profile Slope Angle (m/m)	Read an Existing File (T or F)	Write a New File (T or F)	Files (T or F)		Storage (T or F)
	Field 1 Cols 1-5	Field 2 Cols 6-15	Field 3 Cols 16-25	Field 4 Cols 26-35	Field 5 Col 40	Field 6 Col 45	Field 7 Col 50	Field 7 Col 55	Field 8 Col 60
10	LS Factor	Area	Length	MxSlope	READ	WRITE	ARC	DAT	ACC.
11	1	.99	300.	"blank for default"	F	F	T	T	F

Where:

- Line 1 is the version identification for AgFlow; simply copy and use it as shown.
- Line 2 through 4 are alphanumeric lines for describing and identifying the data; may use any standard ASCII characters on these three lines but are limited to 80 characters for each line.
- Lines 5 through 7 are path names for the TopAGNPS input files, TopAGNPS output files, and AgFlow output files.
NOTE: these paths are now controlled within TopAGNPS control file and should not be changed here. Leave the default directories (".") as shown.
- Line 8 is only a set of column heading fields for the following line. This line is read but not acted upon. The six columns shown are 8-column, column headings fields, right-justified in their fields. Note that columns 33 through 64 are not shown but are reserved for possible future use.
- Line 9 is only a set of column heading fields for the following line. This line is read but not acted upon. The six columns shown are 8-column, column headings fields, right-justified in their fields. Note that columns 33 through 64 are not shown but are reserved for possible future use.
- Line 10 is only a set of column heading fields for the following line. This line is read but not acted upon. The six columns shown are 8-column, column headings fields, right-justified in their fields. Note that columns 33 through 64 are not shown but are reserved for possible future use.
- Line 11 gives the user control over the default values. The nine 5 or 10-column fields are:
 - Field 1 (cols 1-5, "1" is shown) is the code for which TopAGNPS output slope file is to be used for calculating each rasters LS-factor. A "1" will use the flow vector slope file (FVSLOP.ASC); a "2" will use the terrain slope file (TSLOPE.ASC); and a "3" will use the hydraulic slope file (HSLOPE.ASC). A "blank" defaults to "1". The range is "blank" or integer numbers between 1 & 3 inclusive.
 - Field 2 (cols 6-15, ".99" is shown) is a real number for the maximum drainage area before concentrated flow begins (and accumulated sheet & rill erosion cease). A "blank" defaults to 0.99 ha. The range is a "blank" or any positive number.
 - Field 3 (cols 16-25, "300." is shown) is a real number for the maximum profile length before deposition begins and sheet & rill erosion along this particular profile path ceases. A "blank" defaults to 300 m. The range is a "blank" or any positive number.

4. Field 4 (cols 26-35, "*blank for default*" is shown), which means if this field is left blank, the default will be used) is a real number for the maximum profile slope-angle which is meant to be used along with those structural conservation practices that actually decrease the slope-angle such as terraces. A "blank" defaults to "no limit" which should be used for most alternatives especially for natural conditions. The range is *blank* or any positive number.
5. Field 5 (col 40 only, "F" is shown) is a single alphanumeric character to indicate whether an existing LS-factor file (see line 9) from a previous execution is to be read in ("T", "t", "Y", or "y" will read this input file; "N", "n", "F", or "f" will not). The LS-factor file must exist if it is to be read; otherwise an error message will terminate the processing. A "blank" defaults to "false". Acceptable characters are a "*blank*" or "T, t, Y, or y" for true; and "F, f, N, or n" for false.
6. Field 6 (col 45 only, "F" is shown) is a single alphanumeric character to indicate whether an output LS-factor file is to be written ("T", "t", "Y", or "y" will create this output file; "N", "n", "F", or "f" will not). If a pre-existing LS-factor file exists, it will be overwritten; otherwise it is created. A "*blank*" defaults to "false". Acceptable characters are a "*blank*" or "T, t, Y, or y" for true; and "F, f, N, or n" for false.
7. Field 7 (cols 50 & 55, "T" & "T" are shown) is to request both a new SubWta.arc &/or "SubWta.dat" files with the reassigned cell IDs for the channel rasters. A "*blank*" defaults to "true". Acceptable characters are a "*blank*" or "T, t, Y, or y" for true; and "F, f, N, or n" for false.
8. Field 8 (col 60, "F" is shown) is to request that a direct access file be used to store calculations in lieu of internal arrays. The processing time will increase considerably if this option is chosen. A "*blank*" defaults to "false". Acceptable characters are a "T, t, Y, or y" for true; and "*blank*" or "F, f, N, or n" for false.

If the "AGFCNT.INP" control file is not used, then the default values found in line 11 above are used as the default values.

Input data files:

If executing TopAGNPS on a UNIX/Linux system (case sensitive), the following list is in the expected case for each file.

The following input data files described below are spatial layers. The spatial layers must have the same dimensions with the same header information in each file. All of the filenames must be as specified below.

FLOVEC.ASC (TopAGNPS):

This required spatial input file is the flow vector layer.

FVSLOP.ASC (TopAGNPS):

This required spatial input file is the flow vector slope.

HSLOPE.ASC (TopAGNPS):

This required spatial input file is the hydraulic slope.

RELIEF.ASC (TopAGNPS):

This required spatial input file is the elevation layer.

SUBWTA.ASC (TopAGNPS):

This required spatial input file is the subwatershed (cell ID) layer.

TASPEC.ASC (TopAGNPS):

This required spatial input file is the terrain aspect layer.

TSLOPE.ASC (TopAGNPS):

This required spatial input file is the terrain slope layer.

UPAREA.ASC (TopAGNPS):

This required spatial input file is the drainage network layer.

Output files:

There are currently several output files generated by the program.

Log file:

The log file, “AgFlow.log”, is automatically generated and is a log of the processing as the program continues its execution. Additionally, the log files for “TopAGNPS” will also contain information specific for AgFlow. The log file, as it is being written, is also displayed to the screen. The log file contains processing information as well as error and warning messages.

Output data files:

AgFlow_Cell_Data.csv (csv) & AnnAGNPS_Cell.dat (fixed):

These output files are equivalent files that contain information about the cells.

AgFlow_Cell-Reach_Coordinates.csv:

This output file is a csv-formatted file that contains a list of the cell IDs and the coordinates to each cell's connection to the reach.

AgFlow_Reach_Data.csv (csv) & AnnAGNPS_Reach.dat (fixed):

These output files are equivalent files that contain information about the reaches.

AnnAGNPS_Cell_Data_Section.csv:

This csv-formatted file is formatted for use directly into AnnAGNPS as the "Cell Data" section.

AnnAGNPS_Reach_Data_Section.csv:

This csv-formatted file is formatted for use directly into AnnAGNPS as the "Reach Data" section.

AnnAGNPS_SubWta.dat (fixed):

This fixed-formatted file is a spatial file of the cell IDs - (1 value per record).

AgFlow_Cell_Longest_Flow_Path.asc:

This is a spatial layer file containing the longest flow path for each identified cell ID.

AgFlow_LS_Factor.asc:

This is a spatial layer file containing the LS factor values for all rasters within the watershed boundary.

AnnAGNPS_Cell_IDs.asc:

This is a spatial layer file containing cell IDs.

AnnAGNPS_Reach_IDs.asc:

This is a spatial layer file containing reach IDs.

SUBWTA_REASSIGNED.asc:

If the AGFLOW control file is used with the REASSIGN keyword set to “1”, then this spatial file is produced showing the reassigned subarea values. It is not used or required elsewhere within TopAGNPS.